

ORIGINAL RESEARCH

Transmitter selection and receiver placement for target parameter estimation in cooperative radar-communications system

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Funding information

Municipal Government of Quzhou, Grant/Award Number: 2021D005

Abstract

Target parameter estimation is considered for the cooperative multiple-input multiple-output (MIMO) radar and MIMO communications system. To address the hardware limitation of the radar system, a joint transmitter selection and receiver placement (JTSRP) problem is formulated to minimise the estimation performance benchmarked by the Cramer–Rao bound (CRB), where the transmitters can be selected from a discrete set and the receivers can be deployed over a continuous region. To efficiently solve this mix-integer non-linear programming problem approximately, a genetic algorithm (GA)-based method is proposed. It is shown that the result obtained by the proposed algorithm is close enough to the optimum solution of the JTSRP problem. Numerical examples are presented to analyse the performance of the cooperative system designed by the GA-based JTSRP.

KEYWORDS

Cramer–Rao bound, integrated radar and communications system, receiver placement, transmitter selection

1 | INTRODUCTION

The integrated radar and communications (IRC) system have been widely investigated recently [1–6]. Among several popular IRC prototypes, the coexisting radar and communications system attracts considerable attention [7–9]. Multiple-input multiple-output (MIMO) radar shows improved performance for detection and estimation [10–12]. One of the main issues in the coexisting IRC system is to mitigate the interference from the other system. Instead of treating signals from the other system as interference, it is demonstrated that by cooperating properly to share information with each other, both systems may achieve performance gains [13]. As the cooperation between MIMO radar and MIMO communications within a coexisting system can achieve performance gains for both systems, it is regarded as a subcategory in the development of the IRC system [7, 14]. Figure 1 gives a schematic example for the cooperative MIMO radar and MIMO communications system.

The cooperative MIMO radar and MIMO communications system can be deemed as a hybrid active-passive MIMO radar

network, while both the active and passive MIMO radar system would suffer from the issue on hardware limitation. For radar systems, a popular approach to lower the complexity is to apply the transmitter/antenna selection [15, 16]. In the distributed MIMO radar system, transmit antenna correlation and line-of-sight (LOS) reflectivity levels are used to select the most viable antennas with independent diversity branches and strong LOS reflectors [17]. The Cramer–Rao bound (CRB) is used to formulate a knapsack problem in finding the desired antenna subset [18, 19]. The Fisher information matrix (FIM) is also employed for antenna selection [20]. In a passive MIMO radar system, a combinatorial optimisation problem was formulated [21] to jointly place the transmitter and receiver antennae, both from discrete sets, by convex relaxation. The work in Ref. [22] formulated a partition set covering-based problem, which is solved by a hybrid convex optimisation and greedy dropping algorithm. While the aforementioned antenna selections are obtained by gridding the continuous feasible region, the work in Ref. [23] proposed to augment a continuous variable to the grid-based model allowing for off-the-grid sensor placement.

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FIGURE 1 An example of the cooperative multiple-input multiple-output (MIMO) radar and MIMO communications system

Saving the system complexity has attracted considerable attention in the investigation of IRC systems. There have been studies on reducing computational complexity. The work in Ref. [24–28] use convex relaxation to find the beamformer with low computational complexity. The cross-mode parameter estimation algorithm improving the estimation performance with low computational complexity is investigated in Ref. [29]. The complexity reduction potentials is sought in Ref. [30] using low-resolution analogue-to-digital converters. Besides computational complexity, to further reduce the hardware cost in possibly radio frequency chains and processors etc., researchers put efforts on lowering the number of antenna elements. In the array-based IRC system, sparse arrays induced beamformers with shared aperture is implemented in Ref. [31]. The ℓ_1 -norm based problem is proposed in Ref. [32, 33] to find the antenna selection together with the beamformer with low complexity. Most work for the low-complexity IRC system are for the dual-function system but not for the cooperative system.

This paper considers the low-complexity system design for target parameter estimation in the cooperative system with widely separated antennas. In this case, the radar complexity is mainly determined by the selection of transmitted signals to process at each of the receivers. The CRB is derived to benchmark the cooperative system performance for target location and velocity estimation. A joint antenna selection and receiver placement (joint transmitter selection and receiver placement (JTSP)) problem is formulated, which is shown to be a mix-integer non-linear programming (MINLP) problem. The genetic algorithm (GA) is a power tool for the MINLP, and it is a robust global optimiser [34]. The GA-based algorithms solving the MINLP have been investigated to achieve a higher time efficiency [35], to fit for specific problem structures [36], to accelerate the searching process by the graphics processing unit [37], and to extend to large scale MINLP [38] etc. Thus, a GA-based algorithm is proposed to solve the JTSP problem approximately and efficiently. The performance of the

cooperative system designed by the proposed method is analysed through numerical examples.

Notations: Throughout this paper, we use the following notations. The symbol $\odot$ means the Hadamard product, $\otimes$, the Kronecker product, $(\cdot)^T$ and $(\cdot)^H$, the transpose and conjugate transpose, respectively; $\mathbf{I}_K$ denotes a $K \times K$ identity matrix, $\|\cdot\|_0$, the ℓ_0 -norm and $\text{diag}\{\cdot\}$, the diagonal operator.

2 | SIGNAL MODEL

Assume that in a two-dimensional Cartesian coordinate system, the cooperative radar and communications system owns a MIMO radar system with M_R single antenna transmitters and a MIMO communications system with M_C single antenna transmitters. The baseband signal transmitted from the m_R th, $m_R = 1, \dots, M_R$ radar transmitter and the m_C th, $m_C = 1, \dots, M_C$ communications transmitter is $\sqrt{E_{R,m_R}}s_{R,m_R}(kT_s)$ and $\sqrt{E_{C,m_C}}s_{C,m_C}(kT_s)$, respectively, where T_s is the sampling interval, E_{R,m_R} and E_{C,m_C} denote the transmitted power; all waveforms have been normalised such that $\sum_{k=1}^K |s_{R,m_R}(kT_s)|^2 = \sum_{k=1}^K |s_{C,m_C}(kT_s)|^2 = 1/T_s$, and $M_R + M_C = M$. The n th, $n = 1, \dots, N$ single antenna radar receiver¹ locates at $(x_{r,n}, y_{r,n})$. Suppose there is a target at (x, y) moving with speed (v_x, v_y) . Due to cooperation, the communications and radar signals are assumed to have been perfectly estimated or known to the receivers, so the cooperative system manages to make use of signal returns from both systems to estimate the target parameters [13]. At the k th, $k = 1, \dots, K$ sampling instant at the n th radar receiver, the received signal contributed from all transmitters is [13]

¹It is assumed that the communications part has been well designed to guarantee the required performance, and the main focus of this paper is to optimise the radar performance.

$$r_n[k] = \sum_{m_R=1}^{M_R} \sqrt{E_{R,m_R}} \zeta_{R,nm_R} s_{R,m_R}(kT_s - \tau_{R,nm_R}) e^{j2\pi f_{R,nm_R} kT_s} + \sum_{m_C=1}^{M_C} \sqrt{E_{C,m_C}} \zeta_{C,nm_C} s_{C,m_C}(kT_s - \tau_{C,nm_C}) e^{j2\pi f_{C,nm_C} kT_s} + w_n[k], \quad (1)$$

where the time delay, Doppler frequency shift and target reflection coefficient associated with the radar system are denoted by τ_{R,nm_R} , f_{R,nm_R} and ζ_{R,nm_R} , respectively, while these terms for the communications system are denoted by τ_{C,nm_C} , f_{C,nm_C} and ζ_{C,nm_C} , respectively, and $w_n[kT_s]$ is the clutter-plus-noise. Assume that ζ_{R,nm_R} and ζ_{C,nm_C} from all paths are constant over the observation interval, and they have been estimated through preprocessing [12].

The radio frequency chains and processors in receivers may be expensive. Thus, the receivers, confronted by such hardware limitation, can only process a certain amount of signals. However, M_R and M_C in the cooperative system could be large, and it may be unrealistic to process all their transmitted signals. This gives rise to a vital problem on properly selecting the transmitters to ensure a good target parameter estimation performance.

Let $d_{R,nm_R} \in \{1, 0\}$ be a transmitter selection variable, indicating that the n th receiver selects ($d_{R,nm_R} = 1$) or discards ($d_{R,nm_R} = 0$) the signal transmitted from the m_R th radar transmitter. Similarly, let $d_{C,nm_C} \in \{1, 0\}$ be an equivalent selection variable for the communications signals. Define $\mathbf{d}_{R,n} = [d_{R,n1}, \dots, d_{R,nM_R}]^T$ and $\mathbf{d}_{C,n} = [d_{C,n1}, \dots, d_{C,nM_C}]^T$, then $\mathbf{d}_n = [\mathbf{d}_{R,n}^T, \mathbf{d}_{C,n}^T]^T$ implies the selection strategy of the n th receiver. Assume that the receiver hardware limitation constrains each receiver to process signals from W_n transmitters at most, such that $\|\mathbf{d}_n\|_0 = W_n$. The equivalent received signal of Equation (1) under a selection strategy $\mathbf{d}_n$ for the n th receiver is

$$r_n[k] | \mathbf{d}_n = (\mathbf{d}_{R,n} \odot \mathbf{u}_{R,n})^T \mathbf{s}_{R,n}[k] + (\mathbf{d}_{C,n} \odot \mathbf{u}_{C,n})^T \mathbf{s}_{C,n}[k] + w[k]. \quad (3)$$

Stacking K snapshot observations for $r_n[k]$ leads to

$$\mathbf{r}_n | \mathbf{d}_n = [r_n[1], \dots, r_n[K]]^T = \mathbf{D}_{R,n} \odot \mathbf{U}_{R,n} \mathbf{s}_{R,n} + \mathbf{D}_{C,n} \odot \mathbf{U}_{C,n} \mathbf{s}_{C,n} + \mathbf{w}_n, \quad (4)$$

where $\mathbf{D}_{R,n} = \mathbf{I}_K \otimes (\mathbf{d}_{R,n})^T$, $\mathbf{D}_{C,n} = \mathbf{I}_K \otimes (\mathbf{d}_{C,n})^T$, $\mathbf{U}_{R,n} = \mathbf{I}_K \otimes (\mathbf{u}_{R,n})^T$, $\mathbf{U}_{C,n} = \mathbf{I}_K \otimes (\mathbf{u}_{C,n})^T$, $\mathbf{s}_{R,n} = [(\mathbf{s}_{R,n}[1])^T, \dots, (\mathbf{s}_{R,n}[K])^T]^T$, $\mathbf{s}_{C,n} = [(\mathbf{s}_{C,n}[1])^T, \dots, (\mathbf{s}_{C,n}[K])^T]^T$, and $\mathbf{w}_n = [w_n[1], \dots, w_n[K]]^T$. The observations from all receivers under the selection strategy $\mathbf{d} = [\mathbf{d}_1^T, \dots, \mathbf{d}_N^T]^T$ form

$$\mathbf{r} | \mathbf{d} = [\mathbf{r}_1^T, \dots, \mathbf{r}_N^T]^T = \mathbf{F}_R \odot \mathbf{U}_R \mathbf{s}_R + \mathbf{F}_C \odot \mathbf{U}_C \mathbf{s}_C + \mathbf{w} = \mathbf{\Sigma}_R + \mathbf{\Sigma}_C + \mathbf{w}, \quad (5)$$

where $\mathbf{F}_R = \text{diag}\{\mathbf{D}_{R,1}, \dots, \mathbf{D}_{R,N}\}$, $\mathbf{F}_C = \text{diag}\{\mathbf{D}_{C,1}, \dots, \mathbf{D}_{C,N}\}$, $\mathbf{U}_R = \text{diag}\{\mathbf{U}_{R,1}, \dots, \mathbf{U}_{R,N}\}$, $\mathbf{U}_C = \text{diag}\{\mathbf{U}_{C,1}, \dots, \mathbf{U}_{C,N}\}$, $\mathbf{s}_R = [(\mathbf{s}_{R,1})^T, \dots, (\mathbf{s}_{R,N})^T]^T$, $\mathbf{s}_C = [(\mathbf{s}_{C,1})^T, \dots, (\mathbf{s}_{C,N})^T]^T$, $\mathbf{w} = [\mathbf{w}_1^T, \dots, \mathbf{w}_N^T]^T$ is assumed to be zero mean Gaussian white with known variance matrix $\mathbf{Q} = \sigma^2 \mathbf{I}_{NK}$, and $\mathbf{\Sigma}_R = \mathbf{F}_R \odot \mathbf{U}_R \mathbf{s}_R$ and $\mathbf{\Sigma}_C = \mathbf{F}_C \odot \mathbf{U}_C \mathbf{s}_C$ represent the contribution from the radar and the communications end, respectively.

$$r_n[k] | \mathbf{d}_n = \sum_{m_R=1}^{M_R} d_{R,nm_R} \sqrt{E_{R,m_R}} \zeta_{R,nm_R} s_{R,m_R}(kT_s - \tau_{R,nm_R}) e^{j2\pi f_{R,nm_R} kT_s} + \sum_{m_C=1}^{M_C} d_{C,nm_C} \sqrt{E_{C,m_C}} \zeta_{C,nm_C} s_{C,m_C}(kT_s - \tau_{C,nm_C}) e^{j2\pi f_{C,nm_C} kT_s} + w_n[k]. \quad (2)$$

Let $\mathbf{u}_{R,n} = [\sqrt{E_{R,1}} \zeta_{R,n1}, \dots, \sqrt{E_{R,M_R}} \zeta_{R,nM_R}]^T$, $\mathbf{u}_{C,n} = [\sqrt{E_{C,1}} \zeta_{C,n1}, \dots, \sqrt{E_{C,M_C}} \zeta_{C,nM_C}]^T$, $\mathbf{s}_{R,n}[k] = [s_{R,n1}(kT_s - \tau_{R,n1}) e^{j2\pi f_{R,n1} kT_s}, \dots, s_{R,nM_R}(kT_s - \tau_{R,nM_R}) e^{j2\pi f_{R,nM_R} kT_s}]^T$, and $\mathbf{s}_{C,n}[k] = [s_{C,n1}(kT_s - \tau_{C,n1}) e^{j2\pi f_{C,n1} kT_s}, \dots, s_{C,nM_C}(kT_s - \tau_{C,nM_C}) e^{j2\pi f_{C,nM_C} kT_s}]^T$ then, (2) can be rewritten as

3 | JOINT TRANSMITTER SELECTION AND RECEIVER PLACEMENT

The radar task of the cooperative system is to jointly estimate the target location and velocity gathered into a vector $\boldsymbol{\theta} = [x, y, v_x, v_y]^T$. For certain selection strategy $\mathbf{d}$, the log-likelihood function is

$$L(\mathbf{r} | \boldsymbol{\theta}, \mathbf{d}) \propto -(\mathbf{r} - (\boldsymbol{\Sigma}_R + \boldsymbol{\Sigma}_C))^H \mathbf{Q}^{-1} (\mathbf{r} - (\boldsymbol{\Sigma}_R + \boldsymbol{\Sigma}_C)) - \ln(\det(\mathbf{Q})), \quad (6)$$

where $A \propto B$ mean that A is directly proportional to B . The maximum likelihood (ML) estimate of $\boldsymbol{\theta}$ is

$$\hat{\boldsymbol{\theta}}_{ML} = \arg \max_{\boldsymbol{\theta}} L(\mathbf{r} | \boldsymbol{\theta}, \mathbf{d}). \quad (7)$$

The FIM conditioned on $\mathbf{d}$, detailed in Appendix A, is

$$\mathbf{J}(\boldsymbol{\theta}) | \mathbf{d} = 2\Re \left\{ \left(\boldsymbol{\Xi}_{R,\tau}^H + \boldsymbol{\Xi}_{R,f}^H + \boldsymbol{\Xi}_{C,\tau}^H + \boldsymbol{\Xi}_{C,f}^H \right) \mathbf{Q}^{-1} \right. \\ \left. \times \left(\boldsymbol{\Xi}_{R,\tau} + \boldsymbol{\Xi}_{R,f} + \boldsymbol{\Xi}_{C,\tau} + \boldsymbol{\Xi}_{C,f} \right) \right\}, \quad (8)$$

where $\Re\{\cdot\}$ represents taking the real part of a matrix, $\boldsymbol{\Xi}_{R,\tau} = \mathbf{F}_R \odot \mathbf{U}_R(\partial \mathbf{s}_R / \partial \boldsymbol{\tau}_R)(\mathbf{A}_R)^T$, $\boldsymbol{\Xi}_{R,f} = \mathbf{F}_R \odot \mathbf{U}_R(\partial \mathbf{s}_R / \partial \mathbf{f}_R)(\mathbf{B}_R)^T$, $\boldsymbol{\Xi}_{C,\tau} = \mathbf{F}_C \odot \mathbf{U}_C(\partial \mathbf{s}_C / \partial \boldsymbol{\tau}_C)(\mathbf{A}_C)^T$, $\boldsymbol{\Xi}_{C,f} = \mathbf{F}_C \odot \mathbf{U}_C(\partial \mathbf{s}_C / \partial \mathbf{f}_C)(\mathbf{B}_C)^T$, and the definitions for $\mathbf{A}_R$, $\mathbf{B}_R$, $\mathbf{A}_C$ and $\mathbf{B}_C$ are given in Equation (18)–(21) in the appendix.

The CRBs for the estimation of elements of $\boldsymbol{\theta}$ are $\text{CRB}_x = [\mathbf{J}^{-1}(\boldsymbol{\theta})]_{1,1}$, $\text{CRB}_y = [\mathbf{J}^{-1}(\boldsymbol{\theta})]_{2,2}$, $\text{CRB}_{v_x} = [\mathbf{J}^{-1}(\boldsymbol{\theta})]_{3,3}$, and $\text{CRB}_{v_y} = [\mathbf{J}^{-1}(\boldsymbol{\theta})]_{4,4}$, where $[\cdot]_{p,q}$ denotes taking the p th row and q th column element of the matrix. For any unbiased estimator $\hat{\boldsymbol{\theta}}$, the mean square error (MSE) matrix satisfies [39]

$$\mathbb{E} \left\{ \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \right) \left(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta} \right)^H \right\} \succeq \mathbf{J}^{-1}(\boldsymbol{\theta}), \quad (9)$$

where $\mathbf{A} \succeq \mathbf{B}$ means $(\mathbf{A} - \mathbf{B})$ is positive semi-definite. Thus, the CRBs can be used as the metric evaluating the cooperative system target parameter estimation performance after the transmitter selection. To describe the overall localisation and velocity estimation performance, the weighted CRB (WCRB) is calculated,

$$\text{WCRB} \triangleq \alpha_x \text{CRB}_x + \alpha_y \text{CRB}_y + \alpha_{v_x} \text{CRB}_{v_x} + \alpha_{v_y} \text{CRB}_{v_y}, \quad (10)$$

where $\alpha_x = w_x A_x$, $\alpha_y = w_y A_y$, $\alpha_{v_x} = w_{v_x} A_{v_x}$, and $\alpha_{v_y} = w_{v_y} A_{v_y}$. The factor $A_{(\cdot)}$ normalises the CRBs for the location and velocity estimation in different base units, while the weight $w_{(\cdot)}$ specifies the emphasis on specific parameters according to user needs. For example, for the case emphasising on localisation, one can set $w_x = w_y = 1/2$ and $w_{v_x} = w_{v_y} = 0$.

For simplicity, assume the communications and radar transmitters are fixed. The FIM in Equation (8) is a function of the transmitter selection variable $\mathbf{d}$ and all receiver locations. Thus, a joint optimisation for the transmitter selection and receiver placement (JTSRP) can be formulated as followed.

$$\min_{\boldsymbol{\beta}} \quad \text{WCRB} \quad (11)$$

$$\text{s.t.} \quad \boldsymbol{\beta} = \{\mathbf{d}^T, x_{r,1}, y_{r,1}, \dots, x_{r,N}, y_{r,N}\}, \quad (11a)$$

$$\mathbf{d} = [\mathbf{d}_1^T, \dots, \mathbf{d}_N^T]^T, \mathbf{d}_n = [\mathbf{d}_{R,n}^T, \mathbf{d}_{C,n}^T]^T, \quad (11b)$$

$$\mathbf{d}_{R,n} = [d_{R,n1}, \dots, d_{R,nM_R}]^T, \quad (11c)$$

$$\mathbf{d}_{C,n} = [d_{C,n1}, \dots, d_{C,nM_C}]^T, \quad (11d)$$

$$d_{R,nm_R} \in \{0, 1\}, m_R = 1, \dots, M_R, \quad (11e)$$

$$d_{C,nm_C} \in \{0, 1\}, m_C = 1, \dots, M_C, \quad (11f)$$

$$\|\mathbf{d}_n\|_0 \leq W_n, \forall n = 1, \dots, N, \quad (11g)$$

$$(x_{r,n}, y_{r,n}) \in \mathcal{D}_{xy,n}, \mathcal{D}_{xy,n} \subset \mathbb{R}^2, \forall n = 1, \dots, N, \quad (11h)$$

where $\|\mathbf{d}_n\|_0 \leq W_n$ restricts each receiver to process W_n transmitted signals at most, and $\mathcal{D}_n$ is a two-dimensional feasible region to place the n th receiver.

In the optimisation variables in $\boldsymbol{\beta}$ in Equation (11), the selection variables d_{R,nm_R} and d_{C,nm_C} take on discrete values, while $(x_{r,n}, y_{r,n})$ are defined in a continuous region. From Equation (10), the objective WCRB in (11) is a non-linear function of $\boldsymbol{\beta}$. Thus, the problem Equation (11) turns out to be an MINLP [40], which is generally difficult to solve. The MINLP belongs to the combinatorial problems [41] and can be effectively solved by GA as introduced in Ref. [42]. The GA mimics the evolutionary principles and chromosomal processing in natural genetics, which can simulate the variables changes. When the problem contains integer/binary constraints, the GA should modify all generated individuals to be feasible with respect to those constraints, and the entire population remains feasible with respect to integer [43]. When the problem is MINLP, many popular GA solvers (e.g. [44, 45]) can offer a convenient way to find the solution. Here we use GA to solve the problem approximately. In initialising, the feasible regions for all variables in $\boldsymbol{\beta}$ are passed to the GA solver. Next the WCRB can be computed from the CRBs under a given $\boldsymbol{\beta}$. Finally, the GA solver is called to find the approximate solution $\boldsymbol{\beta}^* = [\mathbf{d}^*, x_{r,1}^*, y_{r,1}^*, \dots, x_{r,N}^*, y_{r,N}^*]$ to (11). The procedure is summarised in Algorithm 1.

Algorithm 1 GA-based JTSRP algorithm.

Input: SCNR, $w_x, w_y, w_{v_x}, w_{v_y}, W_1, \dots, W_N, \mathcal{D}_1, \dots, \mathcal{D}_N$.
 1 Set the feasible region of $\mathbf{d}$, $x_{r,1}, y_{r,1}, \dots, x_{r,N}, y_{r,N}$ according to Equations (11e), (11f) and (11h).
 2 Compute $\mathbf{J}(\boldsymbol{\theta}) | \mathbf{d}$ using Equation (8).
 3 Compute $\text{CRB}_x, \text{CRB}_y, \text{CRB}_{v_x}$ and CRB_{v_y} .
 4 Derive the WCRB in Equation (10).
 5 Use GA to approximately solve $\arg \min_{\boldsymbol{\beta}} \text{WCRB}$ in Equation (11).

Output: $\boldsymbol{\beta}^* = [\mathbf{d}^*, x_{r,1}^*, y_{r,1}^*, \dots, x_{r,N}^*, y_{r,N}^*]$.

4 | NUMERICAL EXAMPLES

Assume in the cooperative system, there are $M_R = 3$ radar transmitters and $M_C = 3$ communications transmitters. A set of frequency spread pulsed sinusoidal signals [46] are employed by the radar transmission, whose baseband waveform is $s_{R,m_R}[k] = 1/\sqrt{T} \exp\{j2\pi m_R \Delta f_R k T_s\}$, where Δf_R is the frequency offset between adjacent radar transmitters, and T is the pulse width. The communications system uses the orthogonal frequency-division multiplexing signals [47], whose baseband waveform $s_{C,m_C}[k] = \sum_{i=-\infty}^{\infty} s_{m_C i} (kT_s - iT')$ is composed of $s_{m_C}[kT_s] = \sum_{n=-N_a/2}^{N_a/2-1} a_{m_C}[n] e^{j2\pi n \Delta f k T_s} p_{T'}(kT_s)$, where $a_{m_C}[n]$ is the data symbols, $p_{T'}(kT_s)$ is the rectangular pulse with unit amplitude and width T' , Δf is the frequency space between successive subcarriers, and N_a is the number of subcarriers. Configurations of the radar and communications signals are $\Delta f_R = 1\text{KHz}$, $T = 85\text{ms}$, $\Delta f = 125\text{Hz}$, $N_a = 6$, $T' = 10\text{ms}$, and $E_{R,1} = \dots = E_{C,M_C} = E$. The locations of the two radar transmitters RTx1, RTx2 and RTx3 are $(-52, 376, -46, 440)\text{m}$ $(52, 376, 46, 440)\text{m}$ and $(69, 994, 879)\text{m}$, respectively. The locations of the two communications transmitters CTx1, CTx2 and CTx3 are $(0, 0)\text{m}$ $(-23, 220, 26, 188)\text{m}$ and $(34, 995, 549)\text{m}$, respectively. Assume the cooperative system has $N = 2$ radar receivers. The receiver hardware limitation is the same for all receivers, such that $W_1 = W_2 = W$. Let $W = 4$, and the receiver feasible regions are $\mathcal{D}_{xy,1} = \mathcal{D}_{xy,2} = \mathcal{D}$.

Two types of receiver feasible regions are used, where the first region is composed of N_p discrete points $\mathcal{D}_1 = \{(\rho \cos \varphi_n, \rho \sin \varphi_n) \mid \rho = 70\,000, \varphi_n = 2\pi(n-1)/N_p + 5\pi/12, n = 1, \dots, N_p\}$, and the second region is a continuous set $\mathcal{D}_2 = \{(\rho \cos \varphi, \rho \sin \varphi) \mid \rho \in [0, 70\,000], \varphi \in [0, 2\pi)\}$. Suppose the target presents at $(50.01, 30.26)\text{m}$ with speed $(20, 20)\text{m/s}$, the target reflection coefficient is $0.6 + 0.8j$. The variance of the clutter-plus-noise is $\sigma^2 = 0.01$. Define the signal-to-clutter-plus-noise ratio (SCNR) as $\text{SCNR} = 10 \log_{10}(ME/\sigma^2)$. Set the user specified weights to $w_x = w_y = w_{v_x} = w_{v_y} = 1/4$. The system setup is illustrated in Figure 2.

4.1 | Verification of the CRBs

First, we verify the correctness of the derived CRB. Suppose in the system setup in Figure 2, the points Rx1= $(18, 117, 67\,614)\text{m}$ and Rx2= $(-67, 614, -18, 117)\text{m}$ are chosen to place the receivers. Arbitrarily allocate the receivers using $\mathbf{d}_1 = [1, 1, 1, 1, 0, 0]^T$, $\mathbf{d}_2 = [1, 1, 0, 1, 1, 0]^T$, which indicates that Rx1 selects all radar transmitters and CTx1, and Rx2 selects RTx1, RTx2, CTx1 and CTx2. The RMSEs of the ML estimates and the corresponding CRBs are compared in Figure 3. The ML estimator is asymptotically unbiased, whose variance can be numerically derived from Equation (7). The CRB is calculated from Equation (8), which is the achievable lower bound of any unbiased estimators. When SCNR approaches infinity, the ML estimates can attain the CRB asymptotically. By comparing the RMSE of the ML estimates and the corresponding CRBs, it can examine the correctness of the CRB [39]. Figure 3 shows that the RMSEs

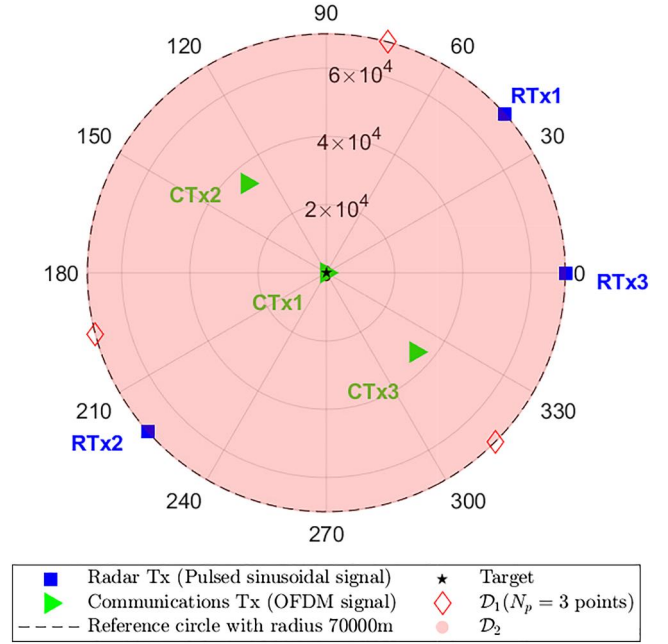


FIGURE 2 System setup

of the location and velocity estimation asymptotically approach the corresponding CRBs, which verifies the correctness of the CRB. In the following numerical example, only the WCRBs, calculated from the CRB, are presented as the estimation performance metric for the cooperative system.

4.2 | JTSRP over a discrete feasible region

With the scenario in Figure 2, we now look at the JTSRP problem over the discrete receiver feasible region $\mathcal{D} = \mathcal{D}_1$, where the set has $N_p = 50$ and $N_p = 300$ discrete points. In both cases, the transmitter selection and the receiver feasible regions are discrete. Algorithm 1 is used to solve the JTSRP problem in Equation (11), where the GA solver is employed to find the solution. There are more than four variables to be optimised, and the GA population size is set to 200 with a max generations 300. For comparison, exhaustive search is performed to find the optimum solution of problem in Equation (11). The normalised WCRBs of the result obtained by Algorithm 1 and the exhaustive search are illustrated in Figure 4. It seems that the GA-based method can achieve almost the same WCRB as the exhaustive search, which shows that the GA-based algorithm can approximately solve the JTSRP problem. Additionally, the GA-based method is faster than the exhaustive search. In the sequel, we focus on GA-based algorithm only to solve the JTSRP problem approximately.

4.3 | JTSRP over a continuous receiver feasible region

This section examines the JTSRP problem over a continuous receiver feasible region $\mathcal{D} = \mathcal{D}_2$ in the system setup in Figure 2.

FIGURE 3 The Cramer–Rao bound (CRB) versus the root-mean-square errors (RMSE) of the maximum likelihood (ML) estimate for a given transmitter selection and receiver placement

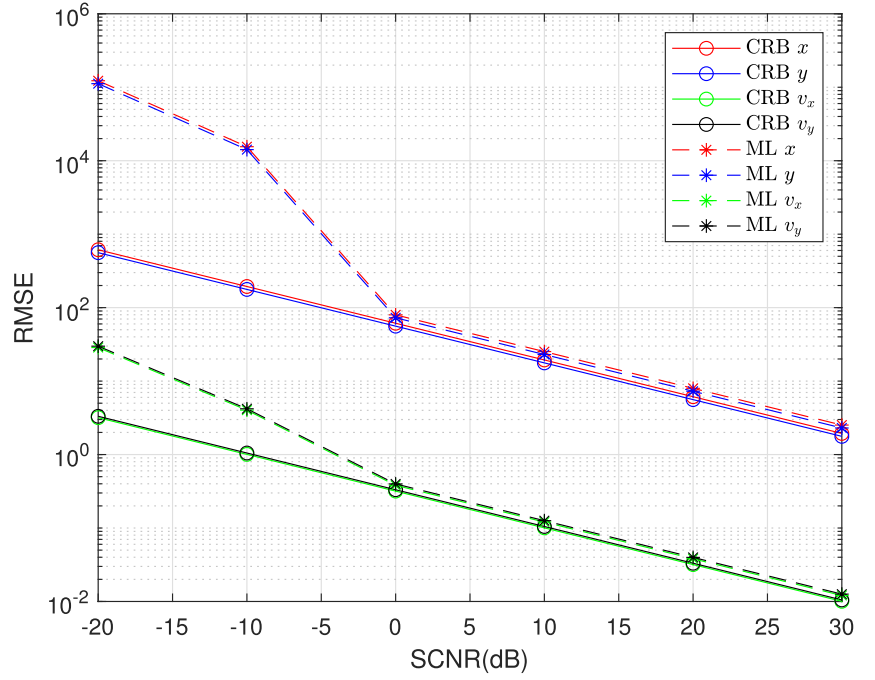
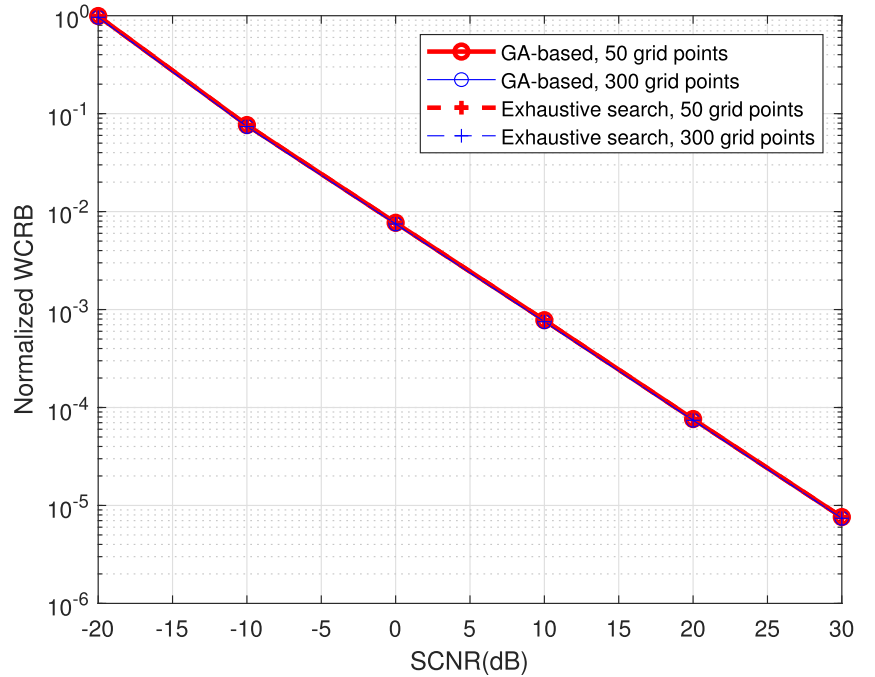


FIGURE 4 The normalised WCRB obtained using exhaustive search and the GA-based method



In this case, the search space is infinite. It is difficult to perform the exhaustive search, so we resort to the GA-based method in Algorithm 3 to solve the problem.

Solutions of the GA-based JTSRP are given in Table 1. It shows that the antenna selections and receiver placements vary among different SCNR regions. Although $W = 4$, the results by JTSRP show that receiver RX1 only picks three transmitted signals, while receiver RX2 only picks one transmitted signals at $\text{SCNR} = -10\text{dB}$. It is noticed that in this case, since the first radar transmitter RTx1 is not elected by any receivers, it can be

turned off. It indicates that the JTSRP can guide the coexisting system designs to widely decrease the system complexity by reducing the radar transmissions and/or receiver processing. To examine the computation complexity of Algorithm 1, in the scenario in Figure 2, the runtimes of Algorithm 1 are calculated for various system configurations. We randomly generate M transmitters (for simplicity, let $M_R = M_C$) and let $N = 2$. Using a personal computer platform with Intel i9-10900 K central processing unit at 3.70 GHz and 128G RAM, Table 2 compares the runtime of Algorithm 1 with the change of the total

transmitter number M and the hardware limitation W at $\text{SCNR} = 10\text{dB}$. Notice that selecting W transmitters among M candidates implies that $W \leq M$. Thus, in the table, the cases when $W > M$ are invalid, which are marked by 'Not available'. It shows that when M and W increase, the runtime of Algorithm 1 also increases. For the conventional MIMO radar configurations, for example, $W \leq 8$ and $M \leq 8$, the runtime is within few minutes.

To examine the advantage of the JTSPR, three more cases are considered: 1) arbitrary transmitter selection and receiver placement; 2) transmitter selection only optimisation with fixed receiver placement; 3) receiver placement only optimisation with fixed transmitter selection. Configurations for these cases are

TABLE 1 Solutions of the GA-based JTSPR over a continuous receiver feasible region

SCNR (dB)	-10	0	10	20
$x_{r,1}$ (m)	55056.556	28890.060	28897.937	-26004.420
$y_{r,1}$ (m)	-37532.639	60071.006	60067.225	61331.157
$x_{r,2}$ (m)	17828.811	17821.568	5987.496	19340.054
$y_{r,2}$ (m)	66644.538	66646.4759	-68660.073	66226.456
$\mathbf{d}_1$	$[0,0,1,1,0,1]^T$	$[0,0,1,1,0,1]^T$	$[0,0,1,1,0,1]^T$	$[0,0,1,1,0,1]^T$
$\mathbf{d}_2$	$[0,1,0,0,0,0]^T$	$[1,1,0,1,0,0]^T$	$[0,1,1,0,0,0]^T$	$[0,1,1,1,0,0]^T$
CRB_x (m)	116.581	36.866	11.658	3.493
CRB_y (m)	167.549	52.983	16.755	6.366
CRB_{v_x} (m/s)	1.014	0.321	0.101	0.027
CRB_{v_y} (m/s)	1.118	0.354	0.112	0.028

Abbreviations: CRB, Cramer–Rao bound; JTSPR, joint transmitter selection and receiver placement.

TABLE 2 Runtime of Algorithm 1 for SCNR = 10 dB under different M and W

runtime (s)	$M = 4$	$M = 6$	$M = 8$	$M = 10$	$M = 12$	$M = 14$	$M = 16$	$M = 18$	$M = 20$
$W = 2$	286	384	554	809	927	953	1460	1757	2064
$W = 3$	228	736	696	953	1247	1499	1737	1972	2108
$W = 4$	455	538	791	1025	1082	1458	1622	1982	2184
$W = 5$	Not available	534	1081	1277	1350	1530	1610	1903	2140
$W = 6$	Not available	506	928	1060	1179	1551	1815	1959	2304
$W = 7$	Not available	Not available	852	1162	1252	1426	1649	2041	2156
$W = 8$	Not available	Not available	757	1022	1449	1487	1553	2192	2254

Abbreviation: SCNR, signal-to-clutter-plus-noise ratio.

TABLE 3 Configuration of different cases

	$\mathbf{d}_1$	$\mathbf{d}_2$	$(x_{r,1}, y_{r,1})$	$(x_{r,2}, y_{r,2})$
Arbitrary selection and placement	$[0,0,1,1,0,1]^T$	$[0,1,1,0,1,1]^T$	(49 992.05, -41 114.87)m	(-43 436.88, 36 180.37)
Transmitter selection only	To be optimised	To be optimised	(49 992.05, -41 114.87)m	(-43 436.88, 36 180.37)
Receiver placement only	$[0,0,1,1,0,1]^T$	$[0,1,1,0,1,1]^T$	To be optimised	To be optimised

given in Table 3. The resulting normalised WCRBs are plotted in Figure 5. It shows that the JTSPR obtains the lowest WCRB in all the tested cases, as expected.

4.4 Identical transmitter selection at different receivers

Now we consider a simplified case, where all receivers share the same transmitter selection results called identical transmitter selection. When the selection strategy $\mathbf{d}_n$ is identical among all receivers, $\tilde{\mathbf{d}} = \mathbf{1}_N \otimes \tilde{\mathbf{d}}_{bm}$, where $\tilde{\mathbf{d}}_{bm} = [\tilde{\mathbf{d}}_R^T, \tilde{\mathbf{d}}_C^T]^T$, $\tilde{\mathbf{d}}_R \triangleq \mathbf{d}_{R,n} = \mathbf{d}_{R,n'}, \tilde{\mathbf{d}}_C \triangleq \mathbf{d}_{C,n} = \mathbf{d}_{C,n'}, \forall n \neq n' \in \{1, \dots, N\}$. The FIM becomes

$$\mathbf{J}(\boldsymbol{\theta}) \mid \tilde{\mathbf{d}} = 2\Re \left\{ \left(\tilde{\mathbf{\Xi}}_{R,\tau}^H + \tilde{\mathbf{\Xi}}_{R,f}^H + \tilde{\mathbf{\Xi}}_{C,\tau}^H + \tilde{\mathbf{\Xi}}_{C,f}^H \right) \times \mathbf{Q}^{-1} \left(\tilde{\mathbf{\Xi}}_{R,\tau} + \tilde{\mathbf{\Xi}}_{R,f} + \tilde{\mathbf{\Xi}}_{C,\tau} + \tilde{\mathbf{\Xi}}_{C,f} \right) \right\}, \quad (12)$$

where $\tilde{\mathbf{\Xi}}_{R,\tau} = \tilde{\mathbf{F}}_R \odot \mathbf{U}_R (\partial \mathbf{s}_R / \partial \boldsymbol{\tau}_R) (\mathbf{A}_R)^T$, $\tilde{\mathbf{\Xi}}_{R,f} = \tilde{\mathbf{F}}_R \odot \mathbf{U}_R (\partial \mathbf{s}_R / \partial \mathbf{f}_R) (\mathbf{B}_R)^T$, $\tilde{\mathbf{\Xi}}_{C,\tau} = \tilde{\mathbf{F}}_C \odot \mathbf{U}_C (\partial \mathbf{s}_C / \partial \boldsymbol{\tau}_C) (\mathbf{A}_C)^T$, $\tilde{\mathbf{\Xi}}_{C,f} = \tilde{\mathbf{F}}_C \odot \mathbf{U}_C (\partial \mathbf{s}_C / \partial \mathbf{f}_C) (\mathbf{B}_C)^T$, $\tilde{\mathbf{F}}_R = \mathbf{1}_N \otimes (\mathbf{I}_K \otimes \tilde{\mathbf{d}}_R^T)$, and $\tilde{\mathbf{F}}_C = \mathbf{1}_N \otimes (\mathbf{I}_K \otimes \tilde{\mathbf{d}}_C^T)$. And the CRB can be computed accordingly.

Consider the system setup in Figure 2 with the continuous feasible region $\mathcal{D}_2$. By adjusting $w_{(\cdot)}$ in Equation (10), two more cases are considered: Case-1: $w_x = w_y = w_{v_x} = w_{v_y} = 1/4$ (Equal weights); Case-2: $w_x = w_y = 0$, $w_{v_x} = w_{v_y} = 1/2$ (All weights on velocity). The normalised WCRBs for these cases are plotted in Figure 6. It shows that for Case-1, the WCRB obtained using different transmitter selection is much lower than the WCRB obtained using identical transmitter selection in all SCNR regions of interest. For

FIGURE 5 Normalised WCRBs for different optimisation approaches

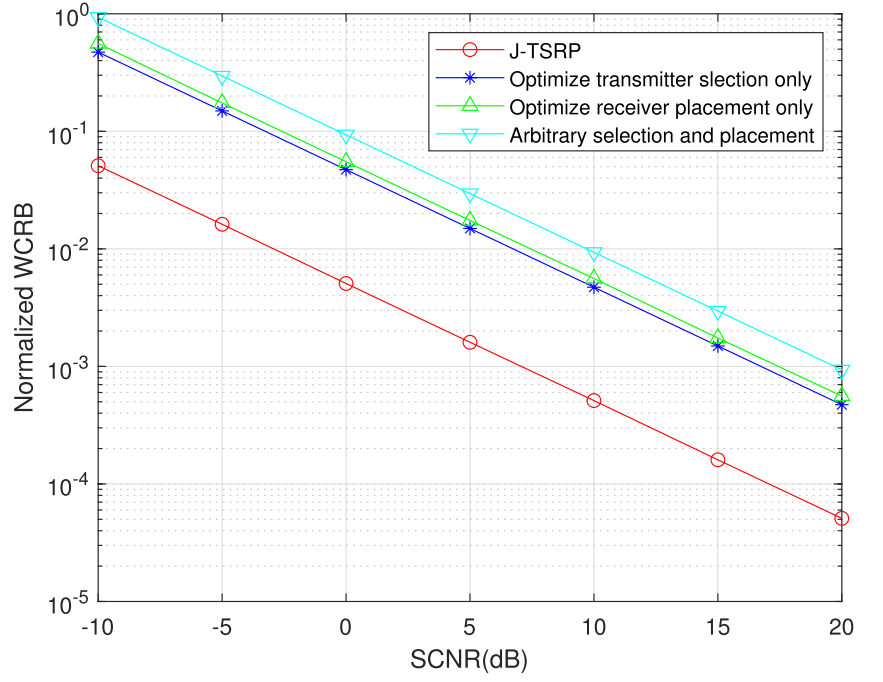
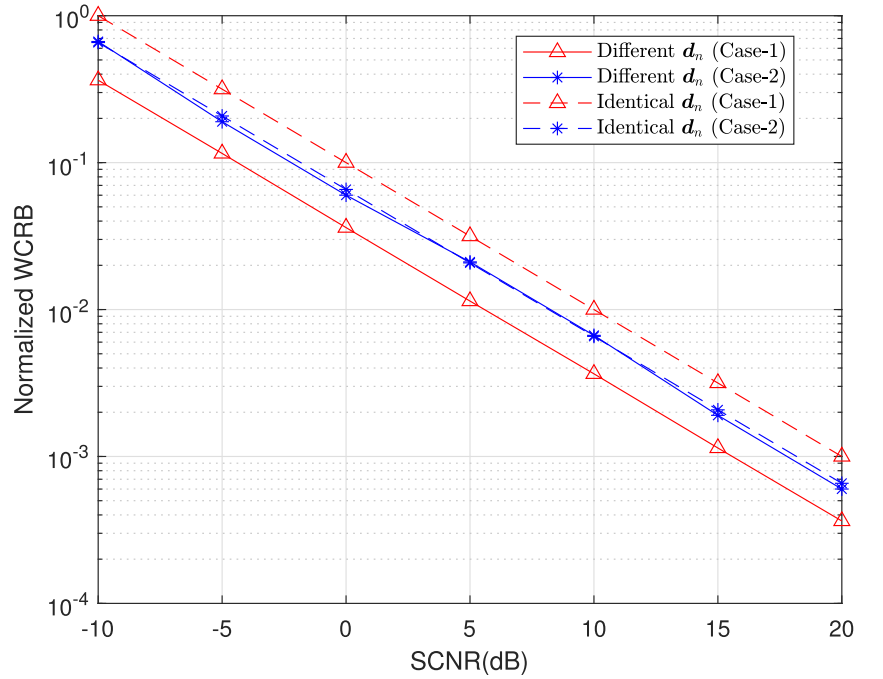


FIGURE 6 The normalised WCRBs using identical and different $\mathbf{d}_n$ among receivers with the joint transmitter selection and receiver placement (JTSRP). Case-1: $w_x = w_y = w_{v_x} = w_{v_y} = 1/4$; Case-2: $w_x = w_y = 0, w_{v_x} = w_{v_y} = 1/2$



Case-2, the WCRB advantage of different transmitter selection becomes slight, but it is still lower than that of the identical transmitter selection. Although the different transmitter selection could lead to complexity in hardware, in the tested cases, it can get a much lower WCRB.

4.5 | Cooperation versus non-cooperation

Next, the estimation performance of the coexisting system using the GA-based JTSRP is studied for the cooperative and

non-cooperative MIMO radar and MIMO communications system. The system setup is the same as Figure 2 with feasible region $\mathcal{D}_2$, and two cases are considered for comparison: Case-1: $w_x = w_y = w_{v_x} = w_{v_y} = 1/4$ (Equal weights); Case-2: $w_x = w_y = 1/2, w_{v_x} = w_{v_y} = 0$ (All weights on location). For the cooperation case, signals from all communications transmitters and the radar transmitters are available for joint optimisation. For the non-cooperation case, the receiver can only use the radar transmitted signals. Figure 7 plots the normalised WCRBs for the cooperative and non-cooperative coexisting system in the two cases designed by the GA-based JTSRP

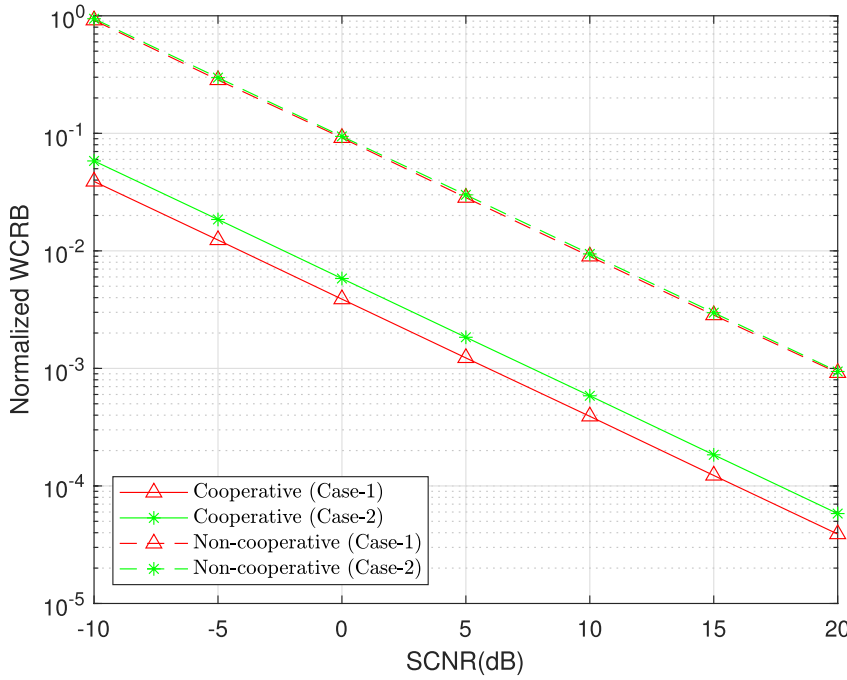


FIGURE 7 The normalised WCRBs for the cooperative and the non-cooperative coexisting system designed by the proposed joint transmitter selection and receiver placement (JTSRP). Case-1: $w_x = w_y = w_{v_x} = w_{v_y} = 1/4$; Case-2: $w_x = w_y = 1/2$, $w_{v_x} = w_{v_y} = 0$

approach. For Case-1, within the SCNR region of interest, the WCRB of the cooperation coexisting system is always lower than the non-cooperation counterpart. This indicates that the JTSRP is able to maintain the estimation performance gain [13] inherited from the cooperative coexisting system. A similar conclusion can be drawn for Case-2.

5 | Conclusion

Target parameter estimation in the cooperative MIMO radar and MIMO communications system was studied. We derived the CRB for the joint localisation and velocity estimation in the presence of radar hardware limitations for the cooperative system. A JTSRP problem was formulated to find the transmitter selection and receiver placement that minimise the WCRB. The problem was solved approximately by a GA-based method. Through numerical examples, we showed that the proposed JTSRP method can help improve the target parameter estimation performance under hardware limitation, compared with the transmitter-selection-only and the receiver-placement-only method. The JTSRP can maintain the performance advantage brought by cooperation over the non-cooperative systems.

ACKNOWLEDGEMENT

This work was supported by the Municipal Government of Quzhou under Grant Number 2021D005.

CONFLICT OF INTEREST

I confirm that the author and all coauthors of this paper have no conflict of interest to disclose.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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How to cite this article: Wang, L., He, Q., Li, H.: Transmitter selection and receiver placement for target parameter estimation in cooperative radar-communications system. *IET Signal Process.* 16(7), 776–787 (2022). <https://doi.org/10.1049/sil2.12108>

APPENDIX A

where ∇_{θ} is a gradient operator, and

The Calculation of $\mathbf{J}(\theta)$

Note that the deterministic unknown target locations and velocities are embedded in the time delays

$$\begin{aligned}\nabla_{\theta}\boldsymbol{\vartheta}^T &= [\mathbf{A}, \mathbf{B}] \\ &= [\mathbf{A}_R, \mathbf{A}_C, \mathbf{B}_R, \mathbf{B}_C],\end{aligned}\quad (17)$$

$$\begin{aligned}\tau_{R,nm_R} &= \frac{\sqrt{(x_{R_t,m_R} - x)^2 + (y_{R_t,m_R} - y)^2} + \sqrt{(x_{r,n} - x)^2 + (y_{r,n} - y)^2}}{c_0}, \\ \tau_{C,nm_C} &= \frac{\sqrt{(x_{C_t,m_C} - x)^2 + (y_{C_t,m_C} - y)^2} + \sqrt{(x_{r,n} - x)^2 + (y_{r,n} - y)^2}}{c_0},\end{aligned}\quad (13)$$

and the Doppler frequency shifts

$$\begin{aligned}f_{R,nm_R} &= \frac{v_x(x_{R_t,m_R} - x) + v_y(y_{R_t,m_R} - y)}{\lambda_R \sqrt{(x_{R_t,m_R} - x)^2 + (y_{R_t,m_R} - y)^2}} + \frac{v_x(x_{r,n} - x) + v_y(y_{r,n} - y)}{\lambda_R \sqrt{(x_{r,n} - x)^2 + (y_{r,n} - y)^2}}, \\ f_{C,nm_C} &= \frac{v_x(x_{C_t,m_C} - x) + v_y(y_{C_t,m_C} - y)}{\lambda_C \sqrt{(x_{C_t,m_C} - x)^2 + (y_{C_t,m_C} - y)^2}} + \frac{v_x(x_{r,n} - x) + v_y(y_{r,n} - y)}{\lambda_C \sqrt{(x_{r,n} - x)^2 + (y_{r,n} - y)^2}},\end{aligned}\quad (14)$$

from all paths, where c_0 is the speed of light, $(x_{R_t,m_R}, y_{R_t,m_R})$ and $(x_{C_t,m_C}, y_{C_t,m_C})$ are the location of the m_R th radar transmitter and the m_C th communications transmitter, respectively, and λ_R, λ_C represent the wavelength of the radar and communications system, respectively. And define an intermediate parameter

$$\begin{aligned}\boldsymbol{\vartheta} &= [\boldsymbol{\tau}^T, \mathbf{f}^T]^T \\ &= [\boldsymbol{\tau}_R^T, \boldsymbol{\tau}_C^T, \mathbf{f}_R^T, \mathbf{f}_C^T]^T,\end{aligned}\quad (15)$$

where $\boldsymbol{\tau}_R = [\tau_{R,11}, \dots, \tau_{R,NM_R}]^T$, $\boldsymbol{\tau}_C = [\tau_{C,11}, \dots, \tau_{C,NM_C}]^T$, $\mathbf{f}_R = [f_{R,11}, \dots, f_{R,NM_R}]^T$, and $\mathbf{f}_C = [f_{C,11}, \dots, f_{C,NM_C}]^T$.

According to Ref. [39], the FIM is

$$\mathbf{J}(\theta) = (\nabla_{\theta}\boldsymbol{\vartheta}^T)\mathbf{J}(\boldsymbol{\vartheta})(\nabla_{\theta}\boldsymbol{\vartheta}^T)^T, \quad (16)$$

which can be partitioned into

$$\mathbf{A}_R = \begin{bmatrix} \frac{\partial \tau_{R,11}}{\partial x} & \dots & \frac{\partial \tau_{R,1M_R}}{\partial x} & \frac{\partial \tau_{R,21}}{\partial x} & \dots & \frac{\partial \tau_{R,NM_R}}{\partial x} \\ \frac{\partial \tau_{R,11}}{\partial y} & \dots & \frac{\partial \tau_{R,1M_R}}{\partial y} & \frac{\partial \tau_{R,21}}{\partial y} & \dots & \frac{\partial \tau_{R,NM_R}}{\partial y} \\ 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix}, \quad (18)$$

$$\mathbf{A}_C = \begin{bmatrix} \frac{\partial \tau_{C,11}}{\partial x} & \dots & \frac{\partial \tau_{C,1M_C}}{\partial x} & \frac{\partial \tau_{C,21}}{\partial x} & \dots & \frac{\partial \tau_{C,NM_C}}{\partial x} \\ \frac{\partial \tau_{C,11}}{\partial y} & \dots & \frac{\partial \tau_{C,1M_C}}{\partial y} & \frac{\partial \tau_{C,21}}{\partial y} & \dots & \frac{\partial \tau_{C,NM_C}}{\partial y} \\ 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \end{bmatrix}, \quad (19)$$

$$\mathbf{B}_R = \begin{bmatrix} \frac{\partial f_{R,11}}{\partial x} & \dots & \frac{\partial f_{R,1M_R}}{\partial x} & \frac{\partial f_{R,21}}{\partial x} & \dots & \frac{\partial f_{R,NM_R}}{\partial x} \\ \frac{\partial f_{R,11}}{\partial y} & \dots & \frac{\partial f_{R,1M_R}}{\partial y} & \frac{\partial f_{R,21}}{\partial y} & \dots & \frac{\partial f_{R,NM_R}}{\partial y} \\ \frac{\partial f_{R,11}}{\partial v_x} & \dots & \frac{\partial f_{R,1M_R}}{\partial v_x} & \frac{\partial f_{R,21}}{\partial v_x} & \dots & \frac{\partial f_{R,NM_R}}{\partial v_x} \\ \frac{\partial f_{R,11}}{\partial v_y} & \dots & \frac{\partial f_{R,1M_R}}{\partial v_y} & \frac{\partial f_{R,21}}{\partial v_y} & \dots & \frac{\partial f_{R,NM_R}}{\partial v_y} \end{bmatrix}, \quad (20)$$

$$\mathbf{B}_C = \begin{bmatrix} \frac{\partial f_{C,11}}{\partial x} & \dots & \frac{\partial f_{C,1M_C}}{\partial x} & \frac{\partial f_{C,21}}{\partial x} & \dots & \frac{\partial f_{C,NM_C}}{\partial x} \\ \frac{\partial f_{C,11}}{\partial y} & \dots & \frac{\partial f_{C,1M_C}}{\partial y} & \frac{\partial f_{C,21}}{\partial y} & \dots & \frac{\partial f_{C,NM_C}}{\partial y} \\ \frac{\partial f_{C,11}}{\partial v_x} & \dots & \frac{\partial f_{C,1M_C}}{\partial v_x} & \frac{\partial f_{C,21}}{\partial v_x} & \dots & \frac{\partial f_{C,NM_C}}{\partial v_x} \\ \frac{\partial f_{C,11}}{\partial v_y} & \dots & \frac{\partial f_{C,1M_C}}{\partial v_y} & \frac{\partial f_{C,21}}{\partial v_y} & \dots & \frac{\partial f_{C,NM_C}}{\partial v_y} \end{bmatrix}. \quad (21)$$

$\mathbf{J}(\boldsymbol{\theta})$ can be computed as [39]

$$[\mathbf{J}(\boldsymbol{\theta})]_{p,q} = 2\Re \left\{ \left(\frac{\partial(\boldsymbol{\Sigma}_R + \boldsymbol{\Sigma}_C)}{\partial \boldsymbol{\theta}_p} \right)^H \mathbf{Q}^{-1} \left(\frac{\partial(\boldsymbol{\Sigma}_R + \boldsymbol{\Sigma}_C)}{\partial \boldsymbol{\theta}_q} \right) \right\}, \quad (22)$$

where the notation $[\cdot]_{p,q}$ denotes taking the p th row and q th column element of the matrix, here we have $p, q = 1, \dots, 2NM$.

When $\boldsymbol{\theta}_p$ rounds over elements in $\boldsymbol{\theta}$, the part $\partial \boldsymbol{\Sigma}_R / \partial \boldsymbol{\theta}$ in Equation (22) is

$$\begin{aligned} \frac{\partial \boldsymbol{\Sigma}_R}{\partial \boldsymbol{\theta}} &= \frac{\partial \mathbf{F}_R \odot \mathbf{U}_R \mathbf{s}_R}{\partial \boldsymbol{\theta}} \\ &= (\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \boldsymbol{\theta}}. \end{aligned} \quad (23)$$

Considering the partitioned structure within $\boldsymbol{\theta}$ in Equation (15), expression Equation (23) can be rewritten as

$$\frac{\partial \boldsymbol{\Sigma}_R}{\partial \boldsymbol{\theta}} = \left[(\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \boldsymbol{\tau}_R}, \mathbf{0}, (\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \mathbf{f}_R}, \mathbf{0} \right]. \quad (24)$$

Plugging Equation (24) and Equation (17) into Equation (16), the right part of Equation (16) becomes

$$\frac{\partial \boldsymbol{\Sigma}_R}{\partial \boldsymbol{\theta}} (\nabla_{\boldsymbol{\theta}} \boldsymbol{\theta}^T)^T = (\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \boldsymbol{\tau}_R} \mathbf{A}_R^T + (\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \mathbf{f}_R} \mathbf{B}_R^T, \quad (25)$$

Similarly, we can derive

$$\frac{\partial \boldsymbol{\Sigma}_C}{\partial \boldsymbol{\theta}} (\nabla_{\boldsymbol{\theta}} \boldsymbol{\theta}^T)^T = (\mathbf{F}_C \odot \mathbf{U}_C) \frac{\partial \mathbf{s}_C}{\partial \boldsymbol{\tau}_C} \mathbf{A}_C^T + (\mathbf{F}_C \odot \mathbf{U}_C) \frac{\partial \mathbf{s}_C}{\partial \mathbf{f}_C} \mathbf{B}_C^T. \quad (26)$$

Eventually, plugging (25) (26) into (16) leads to

$$\begin{aligned} \mathbf{J}(\boldsymbol{\theta}) &= 2\Re \left\{ \left[\mathbf{A}_R \frac{\partial \mathbf{s}_R^H}{\partial \boldsymbol{\tau}_R} (\mathbf{F}_R \odot \mathbf{U}_R)^H + \mathbf{B}_R \frac{\partial \mathbf{s}_R^H}{\partial \mathbf{f}_R} \mathbf{F}_R \odot \mathbf{U}_R^H \right. \right. \\ &\quad \left. \left. + \mathbf{A}_C \frac{(\partial \mathbf{s}_C)^H}{\partial \boldsymbol{\tau}_C} (\mathbf{F}_C \odot \mathbf{U}_C)^H + \mathbf{B}_C \frac{\partial \mathbf{s}_C^H}{\partial \mathbf{f}_C} \mathbf{F}_C \odot \mathbf{U}_C^H \right] \right. \\ &\quad \left. \times \mathbf{Q}^{-1} \left[(\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \boldsymbol{\tau}_R} \mathbf{A}_R^T + (\mathbf{F}_R \odot \mathbf{U}_R) \frac{\partial \mathbf{s}_R}{\partial \mathbf{f}_R} \mathbf{B}_R^T \right. \right. \\ &\quad \left. \left. + (\mathbf{F}_C \odot \mathbf{U}_C) \frac{\partial \mathbf{s}_C}{\partial \boldsymbol{\tau}_C} \mathbf{A}_C^T + (\mathbf{F}_C \odot \mathbf{U}_C) \frac{\partial \mathbf{s}_C}{\partial \mathbf{f}_C} \mathbf{B}_C^T \right] \right\}, \end{aligned} \quad (27)$$

which can be simplified into (8).